



Yiwei Shu

Ph.D. Student in Robotics | Field and Planetary Robotics

📍 Beijing, China

✉️ shuyiwei24@mailsucas.ac.cn

🌐 shu00111.github.io

🔑 Shu00111

Education

Ph.D.	University of Chinese Academy of Sciences , Robotics and Space Robotics - Affiliated with the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences. - Research on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.	Beijing, China 2024 – present
B.Eng.	ShanghaiTech University , Electronic Information Engineering Undergraduate research in robotics, visual control, point-cloud registration, medical imaging, and integrated-circuit design.	Shanghai, China Sept 2020 – June 2024
Exchange	University of Illinois Urbana-Champaign , Exchange Program Completed Quantum Physics, Data Science Discovery, and Statistical Analysis with A+ grades.	Urbana, IL, USA Jan 2023 – May 2023

Manuscripts Under Review and Submitted

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft	June 2026
Yiwei Shu , Shuquan Wang. Manuscript under review at <i>Aerospace Science and Technology</i> . Proposed an IMU/monocular-camera finite-state visual servo controller and validated it with CFD, randomized simulations, and real flight tests; reduced final image-plane radial error from 0.5049 to 0.0027 against a sliding-mode baseline.	
From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation	July 2026
Yiwei Shu et al. Manuscript submitted to <i>IEEE Robotics and Automation Letters</i> (submission status: received). Introduced PostureWaypoint for online configuration-space recovery; evaluated four manipulators across six constrained 3D tasks and validated the method on an RM75 robot.	
RKILPlanner: Risk-based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios	2026
Yiwei Shu , Shuquan Wang. Manuscript submitted to <i>ROBIO 2026</i> . Combined Driver's Risk Field maps, driver-style encoding, and LSTM/CNN features in a differentiable kinematic imitation-learning planner evaluated on the HighD naturalistic highway dataset.	

Publications and Presentations

Expanding Grasp Reachability: Field-based Pose-constrained Two-stage Grasp Planning	July 2026
Yiwei Shu , Shuquan Wang, Jun Li, Hanying Sang, and Yidong Ye. Accepted full paper and oral presentation at <i>ICoSR 2026</i> . Developed reachability and manipulability fields with a two-stage policy that expands pose-constrained grasping near workspace boundaries; validated in simulation and on a 7-DoF RM75 manipulator.	
Learning-based Extended-workspace Manipulation: A Self-relocating Space Crawler for Autonomous Deep-space Outposts	2026
Yiwei Shu , Shuquan Wang. Poster presentation at the <i>46th COSPAR Scientific Assembly 2026</i> . Proposed a passive-rail, self-relocating space crawler and hierarchical reinforcement-learning framework for autonomous extended-workspace manipulation in deep-space outposts.	
Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm	July 2025
Yiwei Shu , Shuquan Wang. Accepted for presentation and publication in the proceedings of the <i>44th Chinese Control Conference (CCC 2025)</i> . Combined visual servoing, adaptive PID control, and Kalman filtering for robust low-cost unpowered-aircraft attitude control.	

Projects

Reliable Autonomous Exploration for Unitree Go2

2026 – present

Reliable active SLAM and navigation for quadruped robots in lunar-analogue and unstructured environments.

- Integrate LiDAR, depth sensing, grid-based mapping, and exploration planning for the Unitree Go2 platform.
- Build Isaac Lab lunar-terrain benchmarks and real-physics evaluation pipelines for legged navigation.
- Investigate calibrated world models, distribution shift, terrain traversability, and safe sim-to-real deployment.

PostureWaypoint for Reactive Whole-Arm Manipulation

2025 – present

Configuration-space waypoint recovery for online manipulation in cluttered and constrained workspaces.

- Resolve end-effector waypoint ambiguity through whole-arm clearance, orientation consistency, and joint-feasibility constraints.
- Evaluate on RM75, Franka Panda, UR5e, and KUKA iiwa manipulators with simulation and real-robot experiments.

Field-based Pose-constrained Two-stage Grasp Planning

2025 – 2026

Reachability- and manipulability-aware grasp planning for objects near manipulator workspace boundaries.

- Construct reachability and manipulability fields directly from the robot model.
- Use a sweep-then-pick policy when direct grasping is infeasible under strict pose constraints.

Autonomous Fixed-Wing Target Landing

2023 – 2026

Low-cost visual-servo control for underactuated and unpowered fixed-wing aircraft.

- Combine CFD-based aerodynamic characterization, IMU guidance, monocular visual servoing, and adaptive control.
- Validate controllers through randomized simulation and physical flight experiments.

Technical Skills

Programming: Python, MATLAB, C, C++, R, Verilog, LaTeX

Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR and depth sensing

Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation

Engineering Tools: Git, Linux, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

Selected Honors

- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)
- Second Prize, RoboMaster Eastern Division (2022)
- Third Prize, RoboMaster National Competition (2022)

Teaching and Leadership

- Teaching Assistant, Digital Circuits, ShanghaiTech University (September-December 2022): led tutorials and assisted with grading.
- Sentry Team Lead, RoboMaster (2021-2023): coordinated embedded programming, PCB design, mechanical modeling, and team planning.